

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 3: Hardware

Subtopic 3.1: Computers and their Components

Concept 3.1.5: Memory Types (RAM, ROM, SRAM, DRAM, PROM, EPROM, EEPROM)

Total Questions: 15

Mark Schemes begin on Page 8

Question 1 | Source: 9618_w25_qp_11 (Q6.c)

(c) State **three** differences between Dynamic RAM (DRAM) and Static RAM (SRAM).

1

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2

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3

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[3]

Question 3 | Source: 9618_s25_qp_11 (Q6.a.ii)

6 A programmer is buying a new computer.

(a) The programmer is considering the following two computers:

Computer 1	Quad-core 2.2GHz processor 16-bit architecture 1 GB Random Access Memory (RAM) 500 GB magnetic hard disk
Computer 2	Dual-core 3.8 GHz processor 32-bit architecture 2 GB RAM 500 GB solid state drive (SSD)

(ii) The two computers have different amounts of RAM.

Explain how different amounts of RAM affect the performance of a computer.

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[3]

Question 4 | Source: 9618_w24_qp_11 (Q3.c)

3 A computer system has a dual-core Central Processing Unit (CPU).

- (c) Identify **two** disadvantages of using Dynamic RAM (DRAM) instead of Static RAM (SRAM) in a computer system.

1

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2

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[2]

Question 5 | Source: 9618_w24_qp_12 (Q2.b)

- 2 Embedded systems are used in many electronic devices.

- (b) An embedded system has Dynamic RAM (DRAM).

Identify **one** benefit of using DRAM instead of Static RAM (SRAM) in an embedded system.

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..... [1]

Question 6 | Source: 9618_w24_qp_12 (Q2.c)

- 2 Embedded systems are used in many electronic devices.

- (c) Give **two** differences between Erasable Programmable ROM (EPROM) and Electrically Erasable Programmable ROM (EEPROM).

Difference 1

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Difference 2

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[2]

Question 7 | Source: 9618_w24_qp_13 (Q2.a)

- 2 A memory buffer uses Dynamic RAM (DRAM).

- (a) Identify **two** differences between DRAM and Static RAM (SRAM).

Difference 1

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Difference 2

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[2]

Question 8 | Source: 9618_s24_qp_12 (Q2.c)

- 2 A computer game is being designed that users will be able to play using a virtual reality (VR) headset.

- (c) The VR headset has Electrically Erasable Programmable Read Only Memory (EEPROM).

Explain the benefits of using EEPROM instead of other types of Read Only Memory (ROM) in the VR headset.

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[3]

Question 9 | Source: 9618_w23_qp_11 (Q7.c)

- (c) A 3D printer contains 1 GB of Dynamic RAM (DRAM) to store print data.

State **two** advantages of the printer having Dynamic RAM instead of Static RAM (SRAM).

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2

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[2]

Question 10 | Source: 9618_w23_qp_13 (Q7.a)

- 7 A laptop computer has Static RAM (SRAM).

A virtual reality headset and a laser printer are connected to the laptop.

- (a) Explain why Static RAM is used in the laptop instead of Dynamic RAM.

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Question 11 | Source: 9618_w23_qp_13 (Q7.b)

- 7 A laptop computer has Static RAM (SRAM).

A virtual reality headset and a laser printer are connected to the laptop.

- (b) Identify **two** reasons for using Electrically Erasable Programmable ROM (EEPROM) in a virtual reality headset.

1

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2

..... [2]

Question 12 | Source: 9618_s23_qp_11 (Q4.a.i)

- 4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

- (a) This computer has both Read Only Memory (ROM) and Random Access Memory (RAM).

- (i) Describe the contents of the ROM in the central computer.

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Question 13 | Source: 9618_s23_qp_11 (Q4.a.ii)

- 4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

(a) This computer has both Read Only Memory (ROM) and Random Access Memory (RAM).

(ii) The central computer has Dynamic RAM (DRAM).

Identify **two** advantages of using DRAM instead of Static RAM (SRAM).

- 1
- 2

[2]

Question 14 | Source: 9618_w22_qp_11 (Q9.b)

9 Many modern televisions are examples of embedded systems.

(b) Embedded systems use Electrically Erasable Programmable ROM (EEPROM).

Describe **one** benefit of using EEPROMs in an embedded system.

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- [2]

Question 15 | Source: 9618_w22_qp_12 (Q3.c)

(c) Embedded systems contain Read Only Memory (ROM) and Random Access Memory (RAM).

Explain the reasons why ROM is used in an embedded system.

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- [2]

Question 16 | Source: 9618_s21_qp_11 (Q5.b)

5 Kiara has a washing machine and a refrigerator.

- (b) The washing machine's embedded system makes use of both Random Access Memory (RAM) and Read Only Memory (ROM).

State the purpose of RAM and ROM within the washing machine's embedded system.

RAM

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ROM

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[2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q6.c)

6(c)	1 mark per bullet point, max 3 marks • DRAM requires to be refreshed/charged SRAM does not request a refresh • DRAM stores each bit as charge SRAM uses flip-flop to store each bit • DRAM is less expensive to manufacture SRAM is more expensive to manufacture • DRAM has slower access speeds SRAM has faster access times • DRAM has higher <u>storage/bit/data density</u> SRAM has lower storage/bit/data density • DRAM is used in main memory SRAM is used in cache	3
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Mark Scheme for Question 2 | Source: 9618_w25_ms_13 (Q4.c)

4(c)

One mark for each correct row

Description	Memory Technology
A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.	EEPROM
A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.	EPROM
A read only memory where the contents can be written only once after manufacture.	PROM

3

Mark Scheme for Question 3 | Source: 9618_s25_ms_11 (Q6.a.ii)

6(a)(ii)	1 mark each to max 3 • More RAM means more currently running data and instructions can be stored • without needing to use virtual memory • without having to fetch the data from secondary storage first • which has a slower access time • Less latency / delay waiting for instructions / data	3
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Mark Scheme for Question 4 | Source: 9618_w24_ms_11 (Q3.c)

3(c)	1 mark for each bullet point (max 2) e.g. • DRAM requires constant refresh cycles unlike SRAM • DRAM has lower access speed than SRAM	2
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Mark Scheme for Question 5 | Source: 9618_w24_ms_12 (Q2.b)

2(b)	1 mark from: • DRAM is less expensive to manufacture/purchase than SRAM in the embedded system • DRAM has a higher bit density per chip // more data can be stored per chip	1
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Mark Scheme for Question 6 | Source: 9618_w24_ms_12 (Q2.c)

2(c)	1 mark for each bullet point (max 2) • EPROM uses ultraviolet light to erase data whilst EEPROM uses an electrical signal to do this • EPROM has to be removed from the circuit board when changing the data whilst EEPROM remains in the circuit when the data is changed • EPROM erases all the data, EEPROM can erase parts of the data	2
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Mark Scheme for Question 7 | Source: 9618_w24_ms_13 (Q2.a)

2(a)	1 mark for each difference e.g. • DRAM requires constant refreshing, but SRAM does not need to be constantly refreshed • DRAM generally has greater capacities per chip, but SRAM generally has lower capacities per chip • DRAM has slower access times than SRAM	2
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Mark Scheme for Question 8 | Source: 9618_s24_ms_12 (Q2.c)

2(c)	1 mark each to max 3 : • EEPROM allows frequent/multiple read/write/erase operations • ... so the headset can take advantage of new features • ... without fully erasing the contents of the firmware in the headset first // can erase a particular byte or the whole EEPROM • ... without removing the chip(s)/firmware from the headset • ... the contents of the firmware in the headset can be changed by the user without technical expertise • Cheaper to manufacture so headset will be cheaper to purchase	3
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Mark Scheme for Question 9 | Source: 9618_w23_ms_11 (Q7.c)

7(c)	1 mark for each bullet point (max 2) • Dynamic RAM has lower cost per unit • A fast access speed is not needed • Higher bit density // more data can be stored per chip	2
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Mark Scheme for Question 10 | Source: 9618_w23_ms_13 (Q7.a)

7(a)	1 mark for each bullet point (max 2). • Static RAM has faster access time • ...because it does not need to be refreshed • ...used on the CPU for improvement of CPU cache speed • Static RAM has lower	2
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Mark Scheme for Question 11 | Source: 9618_w23_ms_13 (Q7.b)

7(b)	1 mark for each bullet point (max 2) • EEPROM allows frequent / multiple read / write / erase operations • ...which means that the contents of the firmware in the headset can be changed easily • ...without fully erasing the contents of the firmware in the headset first	2
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Mark Scheme for Question 12 | Source: 9618_s23_ms_11 (Q4.a.i)

4(a)(i)	1 mark each to max 2 • Stores the bootstrap program // start-up instructions for the central computer // BIOS • Stores the start-up instructions for the CCTV system/cameras // firmware for CCTV • Stores the kernel of the Operating System // stores parts of the Operating System	2
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Mark Scheme for Question 13 | Source: 9618_s23_ms_11 (Q4.a.ii)

4(a)(ii)	1 mark each to max 2 • Costs less per unit • Higher storage density • Simple design – uses fewer transistors	2
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Mark Scheme for Question 14 | Source: 9618_w22_ms_11 (Q9.b)

9(b)	1 mark for each benefit and 1 mark for corresponding expansion (max 2). For example: • no additional equipment is needed to change • ... enables firmware updates by non-technical users • can be erased and reprogrammed several times • ... so firmware can be updated • ... can erase a particular byte or the whole EEPROM • possible to reprogram / update • ... without removing it from the device	2
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Mark Scheme for Question 15 | Source: 9618_w22_ms_12 (Q3.c)

3(c)	1 mark for each bullet point: • to store data that does not change • data must be stored even when device is without power • to store boot up instructions / system software / firmware / BIOS	2
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Mark Scheme for Question 16 | Source: 9618_s21_ms_11 (Q5.b)

5(b)	1 mark for RAM, 1 mark for ROM RAM: • Store the choices/wash program the user has entered // stores the data read from the sensors // stores the time left in the program // by example ROM: • Store the start-up instructions (for the washing cycles)	2
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